



## Oncor Venus Switch to Sam Switch 345-kV Line – ERCOT Independent Review Project Update

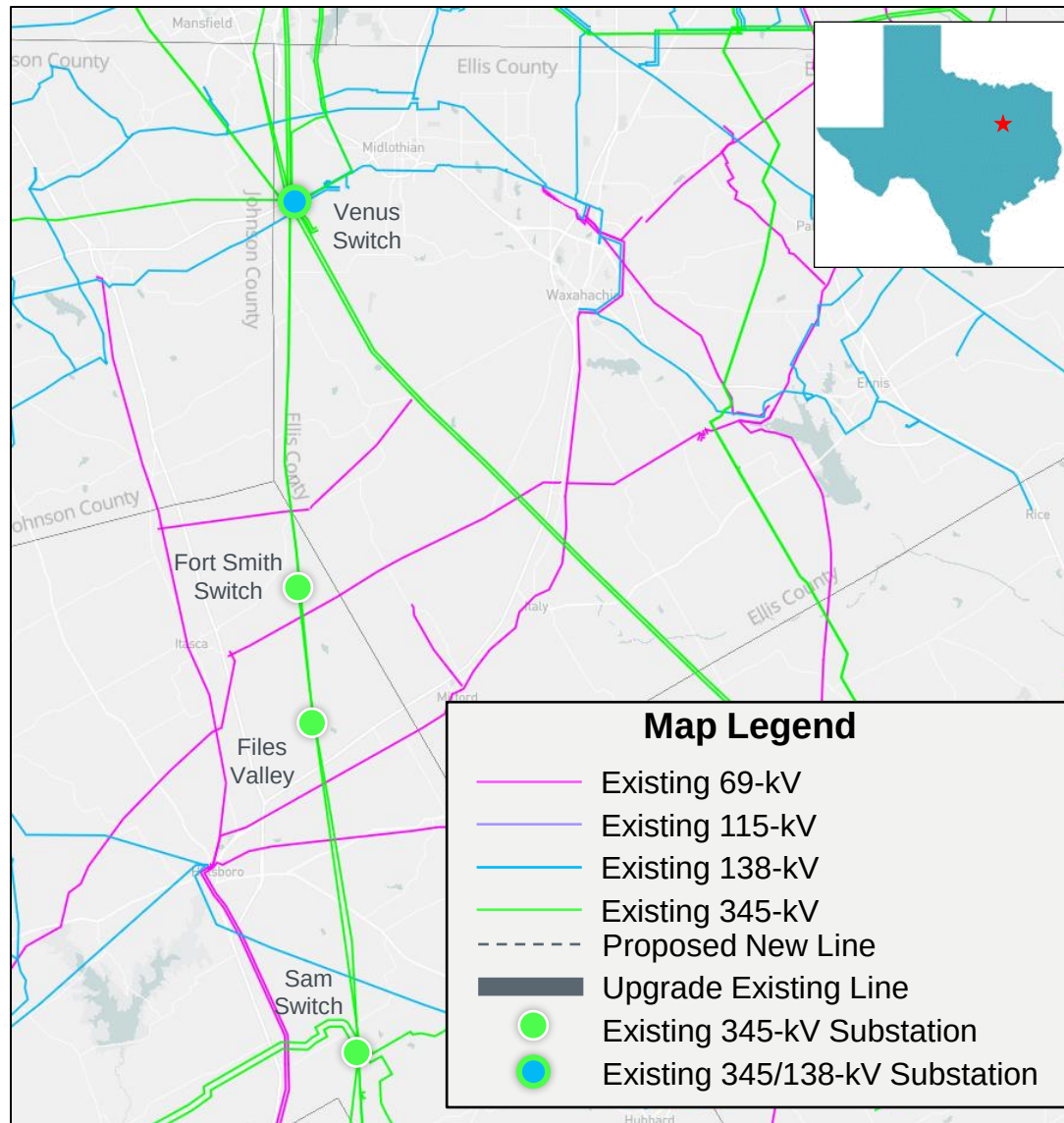
Sarah Gunasekera

RPG Meeting  
December 16, 2024

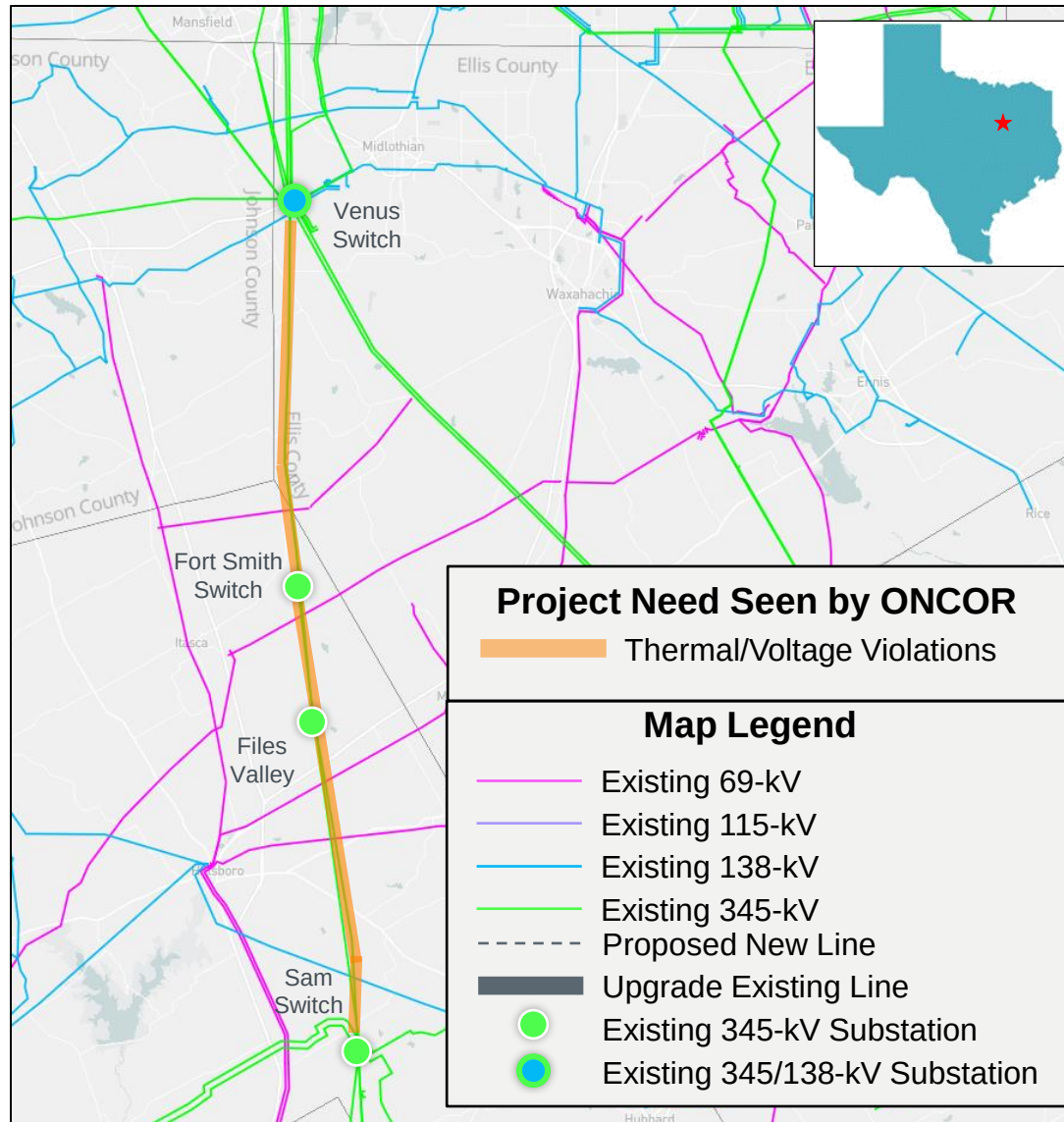
# Introduction

- Oncor submitted the Venus Switch to Sam Switch 345-kV Line Project for Regional Planning Group (RPG) review in June 2024
  - This Tier 1 project is estimated at \$118.9 million and will not require a Convenience and Necessity (CCN)
  - Estimated In-Service Date (ISD) is May 2026
  - Addresses post-contingency thermal overloads on the Venus Switch to Sam Switch double-circuit 345-kV transmission line
- Oncor presented a project overview and ERCOT provided a project scope at the August 2024 RPG Meeting
  - <https://www.ercot.com/calendar/08132024-RPG-Meeting>
- ERCOT provided project updates at the September, October and November RPG Meetings
  - <https://www.ercot.com/calendar/09252024-RPG-Meeting>
  - <https://www.ercot.com/calendar/10162024-RPG-Meeting>
  - <https://www.ercot.com/calendar/11122024-RPG-Meeting>
- This project is currently under ERCOT Independent Review (EIR)

# Recap: Study Area Map



# Recap: Study Area Map with Project Need seen by Oncor



# Recap: Study Assumptions and Methodology Update

- Transmission Updates
  - RPG Projects in the area that have been approved or are being studied with ISDs prior to May 2026 were added to the study base case if not already modeled in the case
    - See Appendix A for the updated list of transmission projects added
  - Transmission projects identified in the 2023 RTP in the study area that have not been approved by RPG were removed
    - See Appendix B for the updated list of transmission projects added
- Generation update
  - Additional 6.9(1) generation was added to based on the July 2024 GIS report
    - See Appendix C for updated list of generation projects added
  - All generation were dispatched consistent with the 2024 RTP methodology
- Loads update
  - Approximately 1,500 MW of confirmed loads in North and North Central Weather Zones were updated to create the study base case

# Recap: Preliminary Results of Reliability Assessment – Updated Base Case

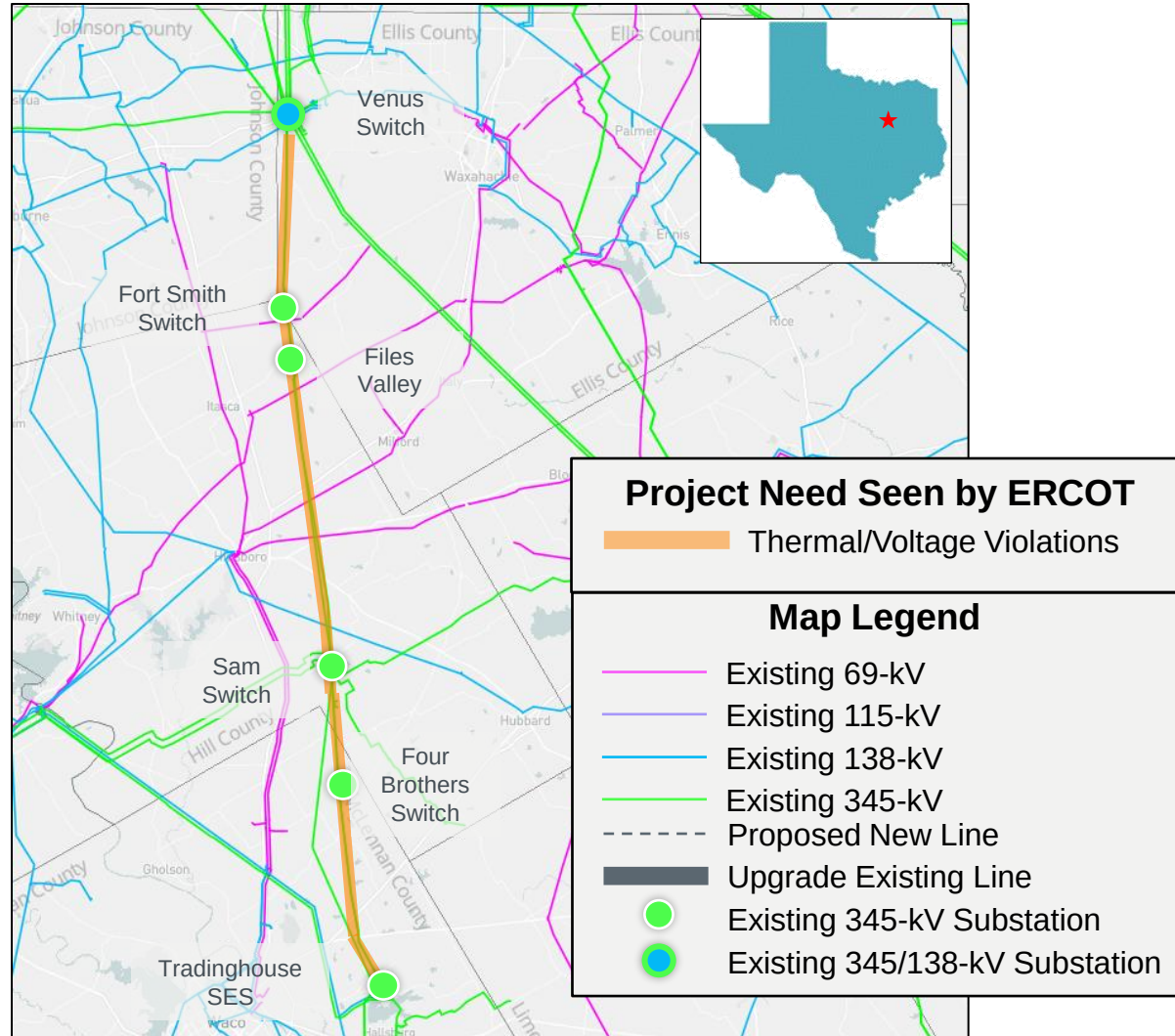
| Contingency Category | Unsolved Power Flow | Voltage Violations | Thermal Overloads |
|----------------------|---------------------|--------------------|-------------------|
| P1                   | None                | None               | 4                 |
| P2, P4, P5           | None                | None               | None              |
| P3<br>(G-1+N-1)*     | None                | None               | 1***              |
| P6.2<br>(X-1+N-1)**  | None                | None               | None***           |
| P7                   | None                | None               | None              |

\*G-1 Generators tested: Comanche Peak SES U1, Midlothian N1, Compadre S1, Sunvalley S1

\*\*X-1 Transformers tested: Sherry Switch X1, Everman Switch X1, Lake Creek SES X1

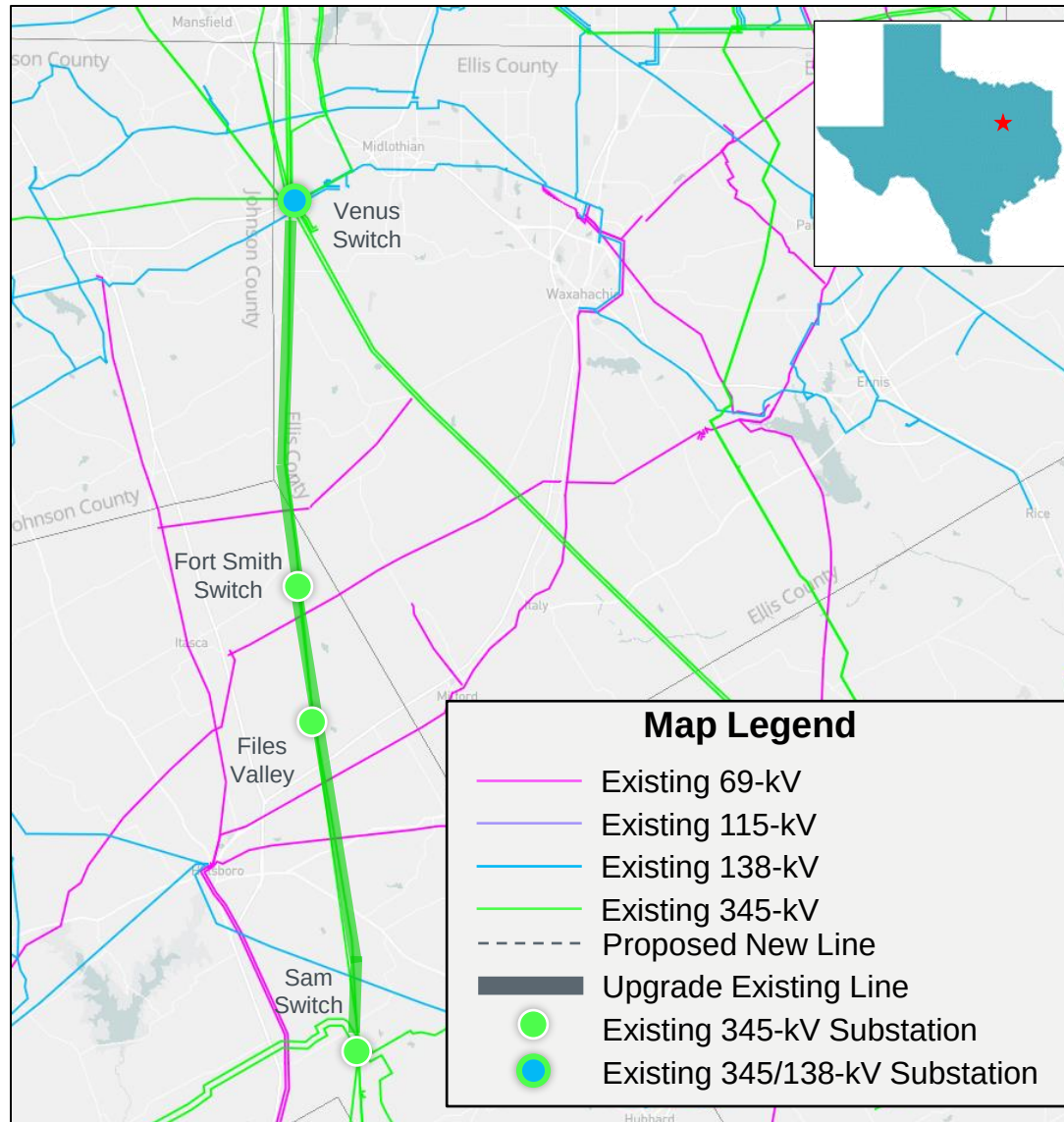
\*\*\*Violations seen in the basecase under P1 events were also seen under G-1 and X-1 events

# Recap: Study Area Map with Updated Project Need seen by ERCOT





# Recap: Option 1 – Oncor Proposed Project

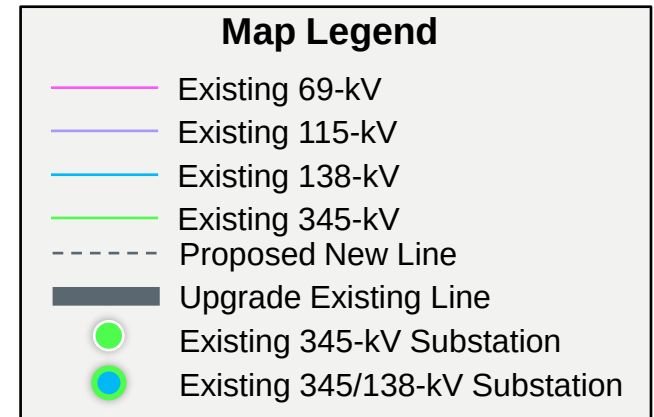
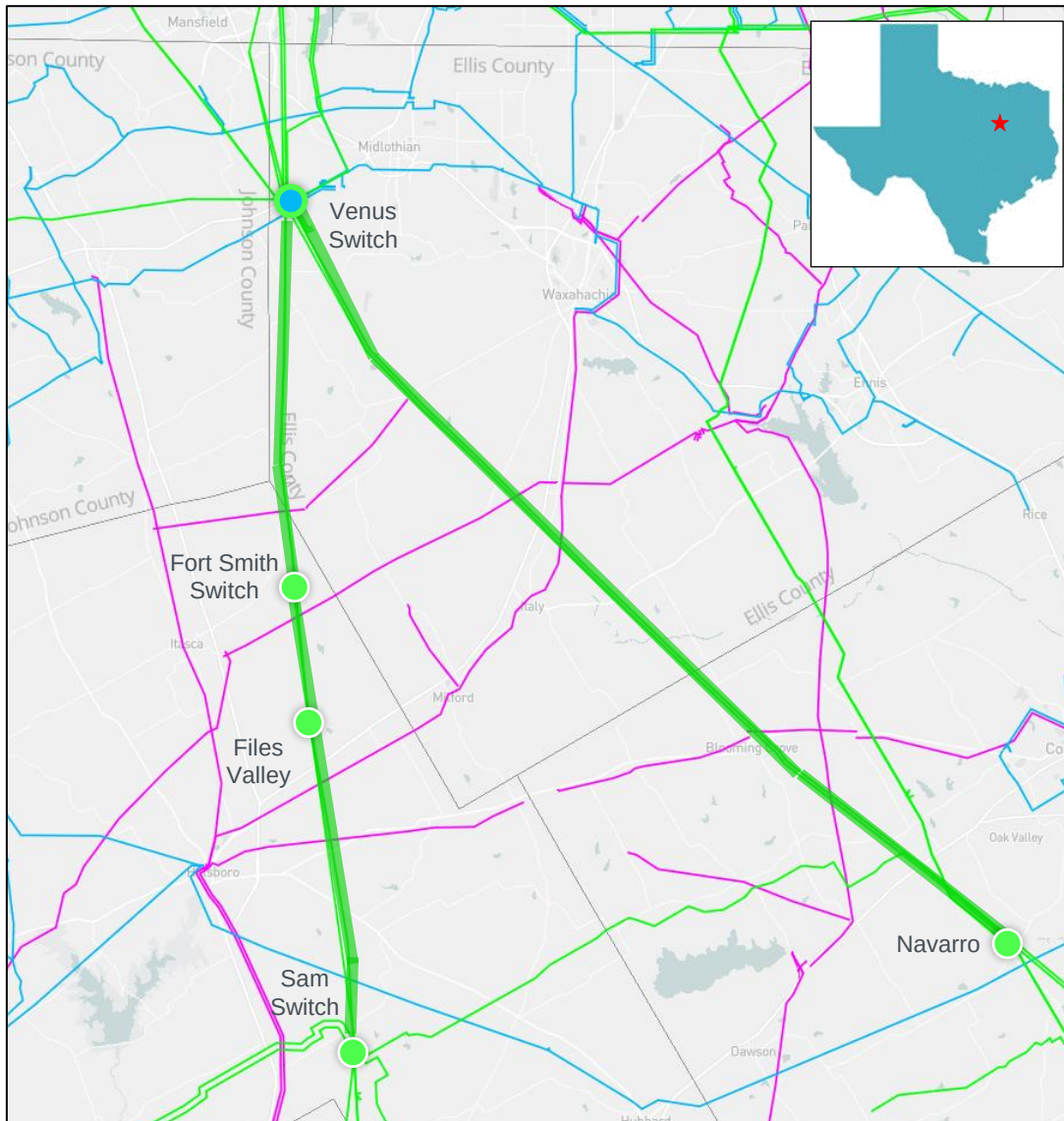




## Recap: Option 1 – Oncor Proposed Project

- Rebuild existing Venus Switch to Fort Smith Switch 345-kV transmission line with normal and emergency ratings of 1912 MVA or greater, approximately 17.8-mile
- Rebuild existing Venus Switch to Sam Switch 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 38.0-mile
- Rebuild existing Fort Smith Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 3.3-mile
- Rebuild existing Sam Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 16.9-mile

# Recap: Option 2 – ERCOT Option



## Recap: Option 2 – ERCOT Option

- Rebuild existing Venus Switch to Fort Smith Switch 345-kV transmission line with normal and emergency ratings of 1912 MVA or greater, approximately 17.8-mile
- Rebuild existing Venus Switch to Sam Switch 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 38.0-mile
- Rebuild existing Fort Smith Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 3.3-mile
- Rebuild existing Sam Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 16.9-mile
- Rebuild existing Venus Switch to Navarro 345-kV double-circuit transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 33.2-mile

# Cost Estimate and Feasibility Assessment

- Transmission Service Providers (TSPs) performed feasibility assessments and provided cost estimates for the options

|          | Cost Estimates<br>(~\$M) | CCN Required<br>(~miles) | Feasibility |
|----------|--------------------------|--------------------------|-------------|
| Option 1 | 118.9                    | 0                        | Yes         |
| Option 2 | 253.5                    | 0                        | Yes         |

# Comparison of Short-Listed Options

|                                            | Option 1  | Option 2  |
|--------------------------------------------|-----------|-----------|
| Addresses Project Needs                    | Yes       | Yes       |
| Meets ERCOT and NERC Reliability Criteria  | Yes       | Yes       |
| Improves Long-Term Load-Serving Capability | Yes       | Yes       |
| CCN Needed                                 | No        | No        |
| Cost Estimate* (\$M)                       | \$118.9 M | \$253.5 M |
| Feasible                                   | Yes       | Yes       |

\* Cost estimates were provided by Transmission Service Providers (TSPs)

- Option 1 is significantly less expensive, compared to Option 2

# ERCOT Preferred Option

- Option 1 was selected as the preferred option because it:
  - Addresses reliability violations and aging infrastructure issue in the study area
  - Is the least cost option and does not require a CCN
  - Improves long-term load-serving capability for future load growth in the area



# Sensitivity Analyses

- Generation Addition Sensitivity Analysis

- ERCOT performed a generation addition sensitivity by adding new the generation listed below to the preferred option case. The additional resources were modeled following the 2024 RTP methodology. ERCOT determined relevant generators do not impact the preferred option

| GINR      | Project Name        | Fuel | Project COD | Capacity (~MW) | County  |
|-----------|---------------------|------|-------------|----------------|---------|
| 21INR0359 | Hickerson Solar     | SOL  | 03/01/2026  | 316.3          | Bosque  |
| 21INR0362 | Oystercatcher Solar | SOL  | 04/15/2026  | 220.3          | Ellis   |
| 24INR0106 | Payne Battlecreek   | SOL  | 05/15/2026  | 85.0           | Hill    |
| 24INR0364 | Pitts Dudik II      | SOL  | 01/29/2026  | 30.2           | Hill    |
| 25INR0018 | Yellow Cat Wind     | WIN  | 03/31/2026  | 300.0          | Navarro |

- Load Scaling Sensitivity Analysis

- ERCOT performed a load scaling sensitivity and concluded that the load scaling did not have a material impact on project need

# Congestion Analysis

- Congestion Analysis

- Congestion analysis was performed for the preferred option using the 2023 RTP 2028 economic case
- The preferred option resulted in one increased congestion within the study area as shown below

| Monitored Line                                                 | % Time of Congestion | New / Existing |
|----------------------------------------------------------------|----------------------|----------------|
| Four Brothers to Tradinghouse SES<br>345-kV transmission line  | 5.27                 | Existing       |
| Four Brothers Switch to Sam Switch<br>345-kV transmission line | 4.22                 | Existing       |

- Upgrading the increased congested line did not yield economic benefit and therefore will not be recommended for upgrade as part of this project.

| Upgrade Tested                                                 | Mileage<br>(~mi) | Passed Production<br>Cost Savings Test | Passed Generation<br>Revenue Reduction Test |
|----------------------------------------------------------------|------------------|----------------------------------------|---------------------------------------------|
| Four Brothers to Tradinghouse SES<br>345-kV transmission line  | 20.2             | No                                     | No                                          |
| Four Brothers Switch to Sam Switch<br>345-kV transmission line | 2.5              | No                                     | No                                          |

# Sub Synchronous Resonance Assessment

- Subsynchronous Resonance (SSR) Assessment
  - Subsynchronous Resonance (SSR) Assessment was conducted for the preferred option per Nodal Protocol Section 3.22.1.3
  - ERCOT found no adverse SSR impacts to the existing and planned generation resources at the time of this study

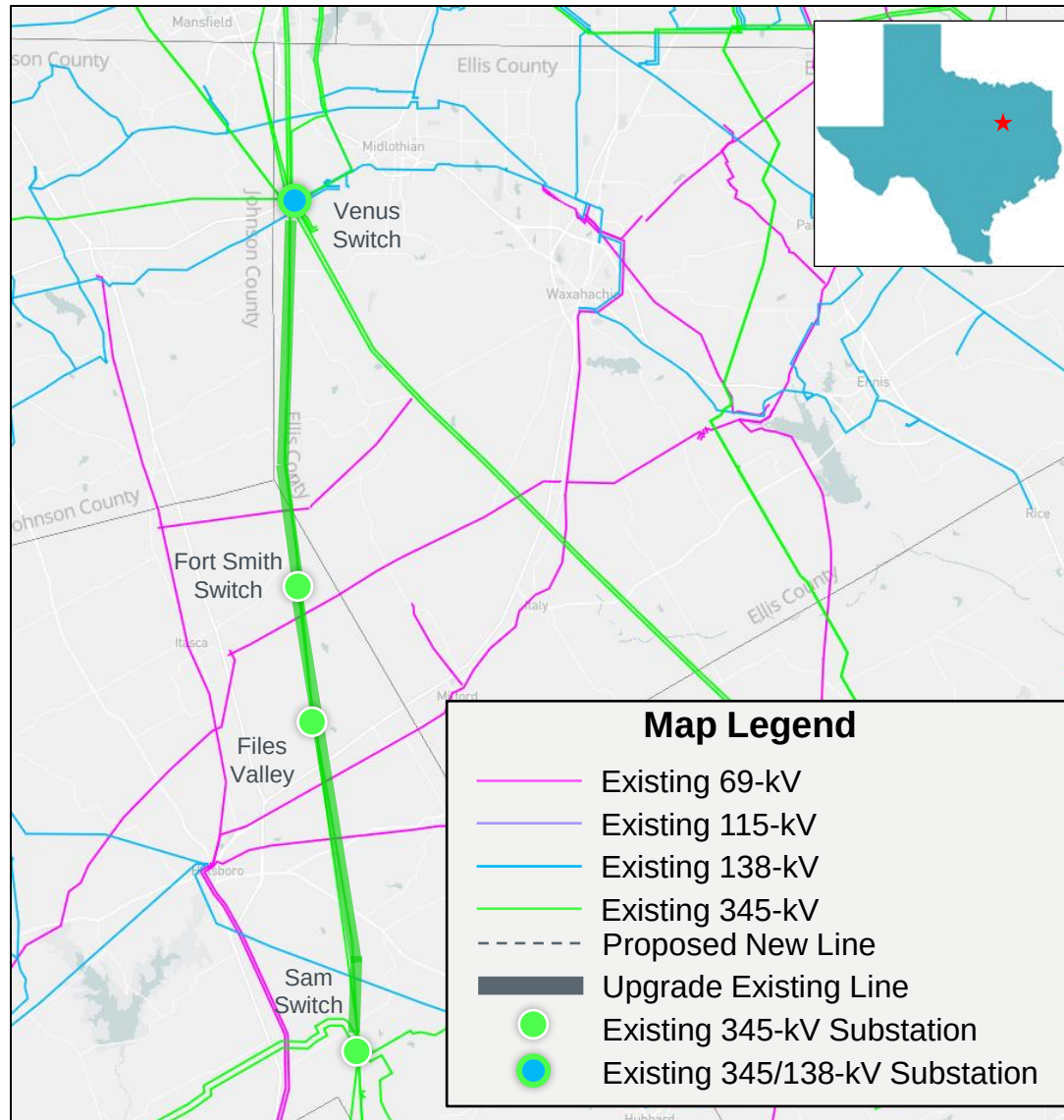
# ERCOT Recommendation

- ERCOT recommends Option 1
  - Estimated Cost: approximately \$118.9 million
  - Expected ISD: May 2026
    - The expected ISD is tentative and are subject to change based on requirements for various approvals, ROW acquisition, and/or construction progress
  - CCN is not required

# ERCOT Recommended Option

- Rebuild existing Venus Switch to Fort Smith Switch 345-kV transmission line with normal and emergency ratings of 1912 MVA or greater, approximately 17.8-mile
- Rebuild existing Venus Switch to Sam Switch 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 38.0-mile
- Rebuild existing Fort Smith Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 3.3-mile
- Rebuild existing Sam Switch to Files Valley 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 16.9-mile

# Map of ERCOT Recommended Option



# Deliverables

- Tentative Timelines
  - EIR report to be posted in the MIS in December 2024
  - EIR recommendation to TAC in January 2025
  - EIR recommendation to R&M in February 2025
  - Seek ERCOT Board of Directors endorsement in February 2025



# *Thank you!*



Stakeholder comments also welcomed through:

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[robert.golen@ercot.com](mailto:robert.golen@ercot.com)

# Appendix A – Transmission Projects Added

| TPIT   | Project Name                                                   | Tier    | Project ISD | County(s)      |
|--------|----------------------------------------------------------------|---------|-------------|----------------|
| 60094  | Convert Waco East - Elm Mott 69 kV Line to 138 kV              | Tier 4  | 5/15/2024   | McLennan       |
| 62666  | Upgrade and convert McGregor - Waco West Line                  | Tier 4  | 12/15/2024  | McLennan       |
| 66216  | Upgrade and convert Waco West - Temple 69 kV Line to 138 kV    | Tier 4  | 6/15/2024   | McLennan, Bell |
| 66218A | Hillsboro - Italy 69 kV Line                                   | Tier 4  | 10/15/2023  | Ellis          |
| 66218B | Hillsboro - Italy 69 kV Line                                   | Tier 4  | 12/15/2025  | Ellis          |
| 71136  | Waxahachie-Waxahachie OCF 69 kV Line Rebuild                   | Tier 4  | 5/15/2025   | Dallas, Ellis  |
| 71903  | Establish Launch Pad 138 kV Switch                             | Tier 4  | 12/15/2025  | McLennan       |
| 72916  | Oncor_N_NoTPIT_Geller 138 kV Substation                        | No TPIT | 12/15/2025  | Dallas         |
| 73443  | Utilize Melton POI via Navarro 345 kV Switch for Project Lefty | Tier 4  | 5/15/2024   | Navarro        |
| 78167  | Add 2nd autotransformer at Trumbull                            | Tier 4  | 11/15/2025  | Ellis          |
| 78367  | Montfort Switch-Shankle Switch 138 kV Line                     | Tier 3  | 12/15/2025  | Navarro, Ellis |
| 80550  | Central Park 138 kV Switch                                     | Tier 4  | 12/15/2024  | McLennan       |
| 82304  | PMCR for adding Blackjack new station                          | Tier 4  | 12/31/2024  | Bosque         |
| 82810  | Olympus 138 kV Switch                                          | Tier 4  | 5/15/2025   | Navarro        |
| 82826  | Sunflower 138 kV Switch                                        | Tier 4  | 5/15/2025   | McLennan       |

# Appendix A – Transmission Projects Added cont'd

| RPG No   | Project Name                                                                | Tier   | Project<br>ISD | County(s)        |
|----------|-----------------------------------------------------------------------------|--------|----------------|------------------|
| 24RPG025 | Gunter 345/138-kV Switch Project                                            | Tier 3 | 12/1/2025      | Collin           |
| 24RPG022 | Wilmer 345/138-kV Switch Project                                            | Tier 1 | 5/1/2026       | Dallas           |
| 24RPG021 | Forney 345/138-kV Switch Rebuild Project                                    | Tier 1 | 12/1/2025      | Kaufman          |
| 24RPG019 | Vineyard Switch to Cypress Waters 138-kV Circuit Addition Project           | Tier 2 | 5/1/2026       | Dallas, Tarrant  |
| 24RPG018 | Salado Switch to Hutto Switch 138-kV Line Project                           | Tier 3 | 5/1/2027       | Bell, Williamson |
| 24RPG001 | Temple Area Project                                                         | Tier 1 | 5/1/2026       | Bell             |
| 23RPG033 | Watermill to Seagoville 138 kV Line Project                                 | Tier 3 | 12/1/2025      | Dallas           |
| 23RPG020 | Hackberry Switch to DFW D East 2 138-kV Double-Circuit Line Section Project | Tier 3 | 12/1/2025      | Dallas           |
| 23RPG018 | Arlington Reliability Enhancement Project                                   | Tier 2 | 5/1/2026       | Tarrant, Dallas  |
| 23RPG006 | North Lake 138 kV Switch Rebuild                                            | Tier 4 | 5/1/2023       | Dallas           |

# Appendix B – Transmission Backed Out

| RTP Project ID | Project Name                                                                                                                        | County(s)       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 2023-NC6       | Telico Area Upgrades                                                                                                                | Ellis           |
| 2023-NC7       | Four Brothers Switch -Tradinghouse - Outlaw - Lake Hall Switch - Sam Switch Area Improvements                                       | McLennan, Ellis |
| 2023-NC13      | Hillboro 138-kV Area Upgrades                                                                                                       | Hill            |
| 2023-NC16      | Sardis Area 138-kV Line Upgrades                                                                                                    | Ellis           |
| 2023-NC19      | Venus - Fort Smith - Sam Switch Double Circuit 345-kV Line Upgrades and Venus Kemp Ranch 345/138-kV Transformer Addition            | Ellis           |
| 2023-NC23      | Venus - Navarro - Jewett Area 345-kV Line Upgrades                                                                                  | Ellis           |
| 2023-NC35      | Navarro (3478) - Haney (213) - Hubbard (3515) 138-kV Line Upgrade                                                                   | Navarro         |
| 2023-NC37      | Hillboro 138/69-kV Transformer Upgrade                                                                                              | Hill            |
| 2023-NC45      | Cleburne Switch (2279) to Keene (2294) to Alvarado (2297) to Griffith (1905) to Railport (442) to Venus (1908) 138-kV Line Upgrades | Johnson         |
| 2023-NC62      | Whitney 345/138-kV Transformer Upgrade                                                                                              | Hill            |

# Appendix C – Generation Added

| GINR      | Project Name                    | Fuel | Project COD | Capacity (~MW) | County   |
|-----------|---------------------------------|------|-------------|----------------|----------|
| 21INR0379 | Ash Creek Solar                 | SOL  | 01/31/2025  | 417.7          | Hill     |
| 23INR0030 | Langer Solar                    | SOL  | 03/01/2027  | 249.8          | Bosque   |
| 23INR0195 | Desert Willow BESS              | OTH  | 02/03/2025  | 154.4          | Ellis    |
| 23INR0349 | Tokio Solar                     | SOL  | 08/25/2025  | 175.7          | McLennan |
| 24INR0023 | Compadre Solar                  | SOL  | 12/25/2024  | 406.1          | Hill     |
| 24INR0038 | SP Jaguar Solar                 | SOL  | 06/01/2026  | 300.0          | McLennan |
| 24INR0039 | SP Jaguar BESS                  | OTH  | 06/30/2025  | 300.0          | McLennan |
| 24INR0138 | Midpoint Storage                | OTH  | 08/30/2025  | 51.3           | Hill     |
| 24INR0139 | Midpoint Solar                  | SOL  | 08/30/2025  | 99.8           | Hill     |
| 24INR0140 | Gaia Storage                    | OTH  | 07/31/2025  | 76.8           | Navarro  |
| 24INR0141 | Gaia Solar                      | SOL  | 07/31/2025  | 152.7          | Navarro  |
| 19INR0110 | Azalea Springs Solar            | SOL  | 05/31/2025  | 181.0          | Angelina |
| 20INR0203 | Pine Forest Solar               | SOL  | 12/01/2025  | 301.5          | Hopkins  |
| 20INR0208 | Signal Solar                    | SOL  | 03/15/2025  | 51.8           | Hunt     |
| 20INR0222 | Tyson Nick Solar                | SOL  | 08/01/2025  | 90.5           | Lamar    |
| 21INR0240 | La Casa Wind                    | WIN  | 03/22/2025  | 148.4          | Stephens |
| 21INR0368 | Eliza Solar                     | SOL  | 12/20/2024  | 151.7          | Kaufman  |
| 21INR0511 | Wolf Ridge Repower              | WIN  | 08/31/2024  | 9.0            | Cooke    |
| 21INR0515 | Roadrunner Crossing Wind II SLF | WIN  | 10/31/2024  | 126.7          | Eastland |
| 22INR0260 | Eliza Storage                   | OTH  | 02/17/2025  | 100.4          | Kaufman  |
| 22INR0526 | Pine Forest BESS                | OTH  | 10/29/2025  | 210.1          | Hopkins  |

# Appendix C – Generation Added

| GINR      | Project Name           | Fuel | Project COD | Capacity (~MW) | County     |
|-----------|------------------------|------|-------------|----------------|------------|
| 22INR0554 | Platinum Storage       | OTH  | 03/03/2025  | 309.5          | Fannin     |
| 22INR0555 | TE Smith Storage       | OTH  | 07/15/2025  | 125.4          | Rockwall   |
| 23INR0026 | Baker Branch Solar     | SOL  | 09/30/2024  | 469.4          | Lamar      |
| 23INR0070 | Chillingham Solar      | SOL  | 10/18/2024  | 352.4          | Bell       |
| 23INR0114 | True North Solar       | SOL  | 12/05/2024  | 238.8          | Falls      |
| 23INR0118 | Blevins Solar          | SOL  | 07/01/2025  | 271.6          | Falls      |
| 23INR0119 | Blevins Storage        | OTH  | 07/01/2025  | 181.3          | Falls      |
| 23INR0296 | Trojan Solar SLF       | SOL  | 02/28/2026  | 151.3          | Cooke      |
| 23INR0367 | Fewell Solar           | SOL  | 09/09/2025  | 203.5          | Limestone  |
| 23INR0403 | Connolly Storage       | OTH  | 09/06/2024  | 125.4          | Wise       |
| 23INR0469 | Big Elm Storage        | OTH  | 11/10/2025  | 100.8          | Bell       |
| 24INR0010 | Pinnington Solar       | SOL  | 10/15/2025  | 666.1          | Jack       |
| 24INR0015 | Five Wells Solar       | SOL  | 09/15/2024  | 322.8          | Bell       |
| 24INR0140 | Gaia Storage           | OTH  | 07/31/2025  | 76.8           | Navarro    |
| 24INR0141 | Gaia Solar             | SOL  | 07/31/2025  | 152.7          | Navarro    |
| 24INR0198 | Two Forks BESS         | OTH  | 07/01/2027  | 309.0          | Cooke      |
| 24INR0295 | Lucky Bluff BESS SLF   | OTH  | 10/15/2025  | 100.8          | Erath      |
| 24INR0312 | Wigeon Whistle BESS    | OTH  | 09/23/2024  | 122.9          | Collin     |
| 24INR0315 | Black Springs BESS SLF | OTH  | 10/15/2025  | 120.7          | Palo Pinto |
| 24INR0631 | Radian Storage SLF     | OTH  | 12/31/2024  | 160.0          | Brown      |
| 25INR0231 | Apache Hill BESS       | OTH  | 11/15/2026  | 201.2          | Hood       |